

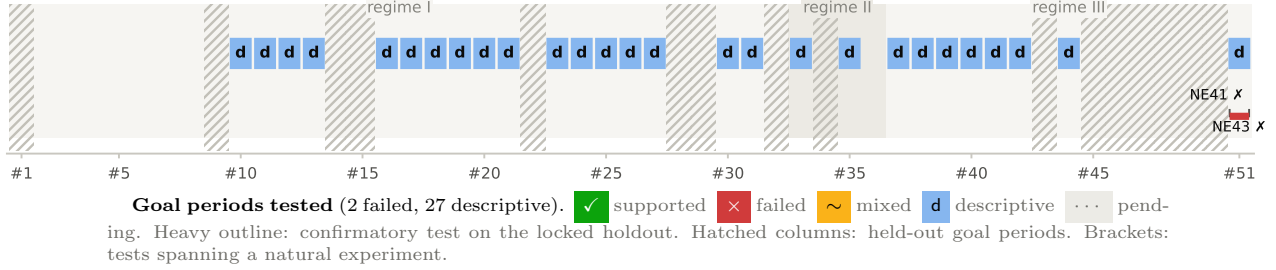
H17: Behavior is a Markov state model with a few metastable sets; its mixing time is a per-period order parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1.5/5

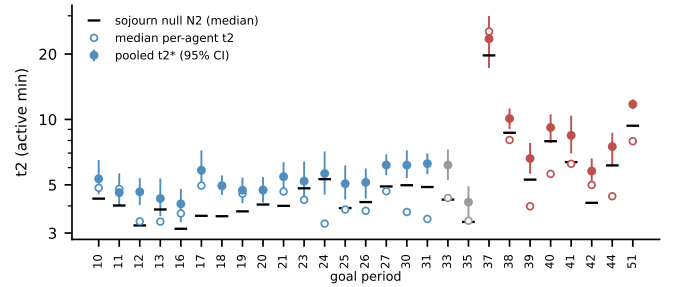
Bottom line. On action classes and on semantic Jev states alike, behavior’s slowest mode is acting vs. waiting plus agent heterogeneity, and its timescale does not order periods by stuckness.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

Is agent behavior a Markov state model (MSM) with a few slow metastable sets, and is its slowest timescale t_2^* a per-period order parameter, slowest in stuck periods? Round 1 fitted MSMs on 6 action classes per active minute in 27 non-holdout periods: CK fails 27/27, t_2^* (5–24 min) barely beats a sojourn null, does not track errors, and regime III is slower than regime I. Verdicts: 13 mixed, 14 failed. Round 1b (improved data) re-ran it on Jev v3.1 behavior states (soft vectors, a noise-cancelling estimator) with real failures instead of stderr. The semantic slowest mode is hours long (agent-day scale), the slow set is still acting vs. waiting, and higher p_{blocked} goes with *shorter* t_2^* . Native tests (forced erasures, nudger switch-off, #27’s spontaneous shift) failed. Holdout not run.



Observables. Slowest implied timescale per goal period at $\tau_c = 5$ min. Filled: pooled MSM (95% agent-day CI). Bars: sojourn-null median (keeps each state’s dwell times, shuffles their order). Open: median per-agent t_2 . The pooled MSM beats the sojourn null in every period, but only modestly, and it sits above the typical agent (agent-mixture inflation). Regime III (red) is slower than regime I (blue). #37’s peak is agents idling before their first and after their last action of the day.

PHYSICAL MODEL

Kinetic Potts / non-equilibrium Markov chain (models 02, 10). State $x_t \in \{\text{browse, type, shell, chat, idle, consolidate}\}$ per agent-minute (H14 scheme); $C(\tau)$: lagged pair counts within agent-days; $T(\tau)$: non-reversible MLE; t_k : implied timescales from eigenvalues λ_k ; χ : PCCA+ memberships (Röblitz-Weber) of m metastable sets; $q^\pm$: forward/backward committers, $\delta = 0$ under detailed balance; p_t : soft (Jev) posteriors.

$$T_{ij}(\tau) = C_{ij}(\tau) / \sum_k C_{ik}(\tau), \quad t_k(\tau) = -\tau / \ln |\lambda_k(T(\tau))|$$

$$t_2^* \equiv t_2(\tau_c = 5 \text{ min})$$

$$\text{CK: } P_{A \rightarrow A}(k\tau) \stackrel{?}{=} [\pi_A^\top T(\tau)^k]_A$$

$$T_{\text{cg}} = (\chi^\top \Pi \chi)^{-1} \chi^\top \Pi T \chi, \quad \delta_i = q_i^+ - (1 - q_i^-)$$

$$\lambda_k^\top = \text{eig}[C(1)^{-1} C(1+\tau)], \quad C(\tau) = \sum_t p_t p_{t+\tau}^\top$$

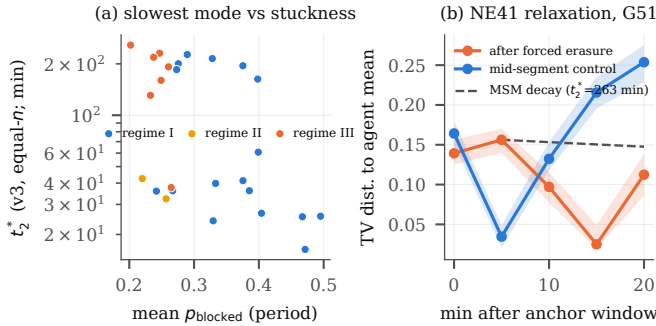
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Candidly: a curiosity with one methodological payoff. The “metastable sets” are tool modality (GUI vs. shell) and idling, which any log viewer could tell you. Part of the GUI-vs-shell split is GUI-heavy vs. shell-heavy agents. t_2^* is a stable, cheap period statistic, but it does not detect stuck periods (no link to error share or output), so an operator gains no monitor. The useful part is negative and transferable: minute-scale agent behavior is strongly non-Markov, pooled MSMs over heterogeneous agents manufacture slow modes, and noisy LLM labels need a noise-cancelling estimator and $\gtrsim 15k$ windows before any timescale means anything. The semantic (Jev) states, the content-level test, show no “stuck” attractor either: waiting is the only slow set.

H17-behavior-metastable-sets · generated 2026-10-04 11:10 by infra/summaries/build_summaries.py · card hypotheses/H17-behavior-metastable-sets/README.md · ratings: H17 re-evaluation agent RE-B1 (suggested; round-1 values were coordinator-calibrated)

RESULTS VS PREDICTIONS

prediction (round 1b)	observed	verdict
P4b t_2^* tracks p_{blocked} (III)	$\rho = -0.69$; stratified -0.49 , p 0.03	(27) failed (sign)
P4b within periods	meta $\rho = -0.28$, p 0.03	(79 failed (sign) agents)
P4b action t_2^* vs real failures	$\rho = -0.19$ (stderr: -0.43)	failed
p_{blocked} flags stuck all: periods	$\rho(\text{fail}) -0.64$; $\rho(\text{output}) -0.71$	III: mixed
P4c regime I slower	40 vs 205 min, p 0.013	reversed
P3d <i>wait</i> is its own set	5/8 regime III; idle+absent	G51: holds
P8 argmax beats R1	MSM 9/27	failed
N1 NE41 relaxation $\approx \tau \approx 21$ min vs t_2^*	263 (G51)	failed
N2 NE43 slower with- out drive	block-to-block swings $\times 6-7$	failed
N3 #27 change point	p 0.31	failed



Second figure. (a) Semantic t_2^* (equal sample size) against the period’s mean p_{blocked} : bimodal (tens of minutes or a day length) and, if anything, shorter where agents are more blocked. (b) After forced erasures (G51) the state deviation decays in ~ 20 min; the MSM’s t_2^* predicts the dashed line. The control rises because of censoring selection.

GOAL PERIODS IN DETAIL

G10 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 G11 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 G12 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 G13 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 G16 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 G17 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 G18 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 G19 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 G20 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 G21 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
 +19 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Synthetic checks (round 1b) show the soft-state shifted estimator is biased $\times 1.6-2.7$ at a 4-h period’s 3,600 windows and breaks down at 12 states below $\sim 15k$ windows; the soft CK has no power and a sojourn null on noisy argmax runs is invalid, so v3 per-period verdicts are descriptive. t_2^* comparisons therefore use equal-size subsamples. The NE41 control anchors are selected (active, censored). p_{blocked} means “waiting on others” in regime I and “failing/looping” in regime III. Pooled MSMs mix agents (R2).

ROUND 2

A hidden-Markov model with agent-day random effects (separates the mixture from switching); content-level states for spirals (#45 is held out).

- **Essence.** What are the metastable cognitive modes (planning, building, debugging, existential talk), and what switches them?
- **H17-R1** A soft Jev-state MSM recovers slow cognitive modes, and switches are driven by external input.
- **H17-R2** Debugging is the slowest mode and the main sink of time.
- **H17-R3** Mode switches synchronize across agents only through shared artifacts (E3).